

Coffee Shop Sales Analysis Dashboard

Project Objective

The primary objective of this project is to analyze coffee shop sales data and generate actionable insights that can improve business performance, customer satisfaction, inventory planning, and sales strategies. The dashboard provides an interactive visualization of sales trends, customer behavior, product performance, and store-wise performance.

1. Overall Business Performance (KPI Cards)

The dashboard begins with key performance indicators (KPIs) that provide a quick overview of the coffee shop's performance.

Total Sales

- Total Revenue Generated: **\$698,812.33**
- Represents the total income earned from all transactions during the selected period.

Total Footfall

- Total Customers Served: **149,116**
- Indicates the number of customer visits across all stores.

Average Bill Per Person

- Average Revenue per Customer: **\$4.69**
- Helps evaluate customer spending behavior.

Average Order Per Person

- Average Items Purchased: **1.44**
- Indicates the average quantity of products purchased by each customer.

Business Insight:

These KPIs provide management with an instant understanding of overall sales performance and customer purchasing patterns.

2. Sales Variation by Hour of the Day

Visualization:

QTY Ordered Based on Hours (Line Chart)

Purpose:

Addresses the project question:

"How do sales vary by day of the week and hour of the day?"

Analysis:

- Orders increase sharply between **7 AM and 10 AM**.
- Peak demand occurs around **9–10 AM**.
- Sales gradually decline after morning hours.
- Lowest sales occur near closing time.

Business Insight:

- Morning hours are the busiest period.
 - More staff should be allocated during breakfast hours.
 - Promotions can be introduced during low-demand periods to increase sales.
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3. Peak Sales Activity

Visualization:

Hourly Sales Trend

Purpose:

Answers:

"Are there any peak times for sales activity?"

Analysis:

- Peak activity occurs during morning hours.
- Customers primarily visit coffee shops before work or during breakfast.

Business Insight:

- Ensure sufficient inventory and staff availability during peak hours.
 - Introduce breakfast combo offers to maximize revenue.
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4. Day-Wise Order Analysis

Visualization:

Day-wise Orders (Bar Chart)

Purpose:

Further supports the analysis of sales by day of the week.

Analysis:

- Friday records the highest number of orders.
- Saturday shows comparatively lower order volume.
- Weekdays generally generate stable sales.

Business Insight:

- Higher workforce allocation is required on busy days.
 - Weekend promotional campaigns can increase customer traffic.
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5. Store Location Performance**Visualization:****Footfall Over Various Stores**

Stores:

- Astoria
- Hell's Kitchen
- Lower Manhattan

Purpose:

Answers:

"How do sales vary across different store locations?"

Analysis:

- Hell's Kitchen generates the highest sales revenue.
- Astoria and Lower Manhattan show similar performance.
- Customer footfall is relatively balanced across stores.

Business Insight:

- Best-performing stores can be used as benchmarks.
 - Underperforming stores may require targeted marketing efforts.
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6. Product Category Sales Distribution

Visualization:

Category Percentage Distribution Based on Sales (Pie Chart)

Purpose:

Answers:

"How do sales vary by product category and type?"

Analysis:

- Coffee contributes the largest share of revenue (**39%**).
- Tea contributes approximately **28%**.
- Bakery products account for around **12%**.
- Other categories contribute smaller percentages.

Business Insight:

- Coffee remains the core revenue driver.
 - Inventory planning should prioritize coffee products.
 - Additional promotions can be designed for low-performing categories.
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7. Product Size Distribution

Visualization:

Size Distribution Based on Orders (Pie Chart)

Analysis:

- Regular Size: **31%**
- Large Size: **30%**
- Not Defined: **30%**
- Small Size: **9%**

Business Insight:

- Customers mostly prefer Regular and Large sizes.
- Small-sized products have relatively low demand.
- Pricing and promotional strategies can focus on larger sizes.

8. Best-Selling Products

Visualization:

Top 5 Products Based on Sales (Bar Chart)

Products:

1. Barista Espresso
2. Brewed Chai Tea
3. Hot Chocolate
4. Gourmet Brewed Coffee
5. Brewed Black Tea

Purpose:

Answers:

"Which products are the best-selling in terms of quantity and revenue?"

Analysis:

- Barista Espresso generates the highest revenue.
- Tea-based beverages also contribute significantly.
- Premium coffee products perform strongly.

Business Insight:

- Best-selling products should always remain in stock.
- These products can be used in promotional bundles and upselling strategies.

9. Interactive Filters

Month Filter

Allows users to analyze sales for:

- January
- February
- March
- April

- May
- June

Day Filter

Allows users to analyze specific weekdays.

Purpose:

- Enables dynamic analysis.
- Helps identify seasonal and weekly sales trends.

Business Insight:

Management can make data-driven decisions by focusing on specific months or days.

Conclusion

The Coffee Shop Sales Dashboard effectively answers all business questions outlined in the project requirements. The analysis reveals that:

- Morning hours are the peak sales period.
- Fridays generate the highest order volume.
- Hell's Kitchen is the best-performing store.
- Coffee is the dominant revenue-generating category.
- Barista Espresso is the top-selling product.
- Customers prefer Regular and Large-sized beverages.
- Average customer spending is approximately **\$4.69 per visit**.

These insights can help management optimize staffing, inventory management, product offerings, and marketing strategies, ultimately improving the overall profitability of the coffee shop.